Accelerating Scientific Research Methodology with AI Tools

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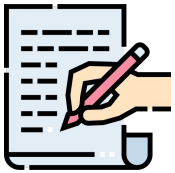
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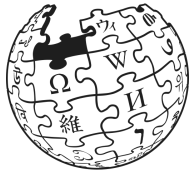
Why academic researcher must incorporate AI tools into their research workflow ?

1. Traditional Research
2. Internet Research
3. Artificial intelligence

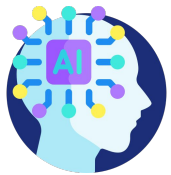














Plan of the presentation

- **Benefits** of using AI tools for academic research.
- **Limitations** of AI tools.
- **Ethical considerations** regarding the use of AI.
- **AI tools** for academic research.
- **ChatDOC** Tutorial.
- **Workshop** : incorporate AI into your research process.
- **Bonus**.

Benefits of using AI tools for academic research

- Brainstorm research topics and questions.
- Come up with new and creative perspectives and approaches to your research that you didn't consider before.
- AI makes knowledge more accessible.
- Work efficiency ⇒ reduce time and effort required for reading, summarizing and writing.
- More accurate translations ⇒ Access to works from diverse backgrounds, countries and languages.
- Critically analyse the strengths and weaknesses of your methodology ⇒ improve the feedback of your reviewers.

Limitations of AI tools

- Lack of genuine understanding.
- Not able to access information beyond training data.
- Human review of AI responses is absolutely necessary.
- Limited creativity ⇒ it's creativity is limited to some patterns, it's difficult to generate truly innovative and creative content.
- Lack of transparency ⇒ it's just a black box !
- Variability in responses. This is an advantage or inconvenient ?
- Data Sensitivity.

Ethical considerations regarding the use of AI

“AI will never replace human, AI will support human”

- You can use AI tools but **you can’t copy and past AI generated text**. You as author, you are fully responsible about the quality, accuracy and originality of your final work.
- You must ensure that the information is correct. **Avoid wrong information !**
- **Be careful about Data Protection Laws**.
- **Understand the limitations of AI tools** ⇒ It will never substitute human critical thinking.
- Stay updated on **the best practices for the use of AI in academia**.

Ressources

- *Christou, Prokopis A. "How to Use Artificial Intelligence (AI) as a Resource, Methodological and Analysis Tool in Qualitative Research?" Qualitative Report 28.7 (2023): 1968-1980.*
- *Eke, Damian Okaibedi. "ChatGPT and the rise of generative AI: Threat to academic integrity?" Journal of Responsible Technology 13 (2023): 100060.*
- *Foltynek, Tomas, et al. "ENAI Recommendations on the ethical use of Artificial Intelligence in Education." International Journal for Educational Integrity 19.1 (2023): 12.*
- *Harker, Jennifer. "Science Journals Set New Authorship Guidelines for AI-generated Text." National Institute of Environmental Health Sciences, 1 Mar. 2023, factor.niehs.nih.gov/2023/3/feature/2-artificial-intelligence-ethics.*
- *Hosseini, Mohammad, Lisa M. Rasmussen, and David B. Resnik. "Using AI to write scholarly publications." Accountability in research (2023): 1-9.*
- *Imran, Muhammad, and Norah Almusharraf. "Analyzing the role of ChatGPT as a writing assistant at higher education level: A systematic review of the literature." Contemporary Educational Technology 15.4 (2023): ep464.*
- *Jarrah, A. M., Y. Wardat, and P. Fidalgo. "Using ChatGPT in academic writing is (not) a form of plagiarism: What does the literature say." Online Journal of Communication and Media Technologies 13.4 (2023): e202346.*
- *Lund, Brady D., et al. "ChatGPT and a new academic reality: Artificial Intelligence-written research papers and the ethics of the large language models in scholarly publishing." Journal of the Association for Information Science and Technology 74.5 (2023): 570-581.*
- *Mijwil, Maad M., et al. "ChatGPT and the future of academic integrity in the artificial intelligence era: a new frontier." Al-Salam Journal for Engineering and Technology 2.2 (2023): 116-127.*
- *Wach, Krzysztof, et al. "The dark side of generative artificial intelligence: A critical analysis of controversies and risks of ChatGPT." Entrepreneurial Business and Economics Review 11.2 (2023): 7-24.*

AI tools for academic research - General AI chatbots

- ChatGPT is a general AI chatbot.
- Using ChatGPT, you can summarize information, get explanations, translate text , etc.
- In the paid version (GPT4), you can even interpret images, pdfs, excel files, etc.



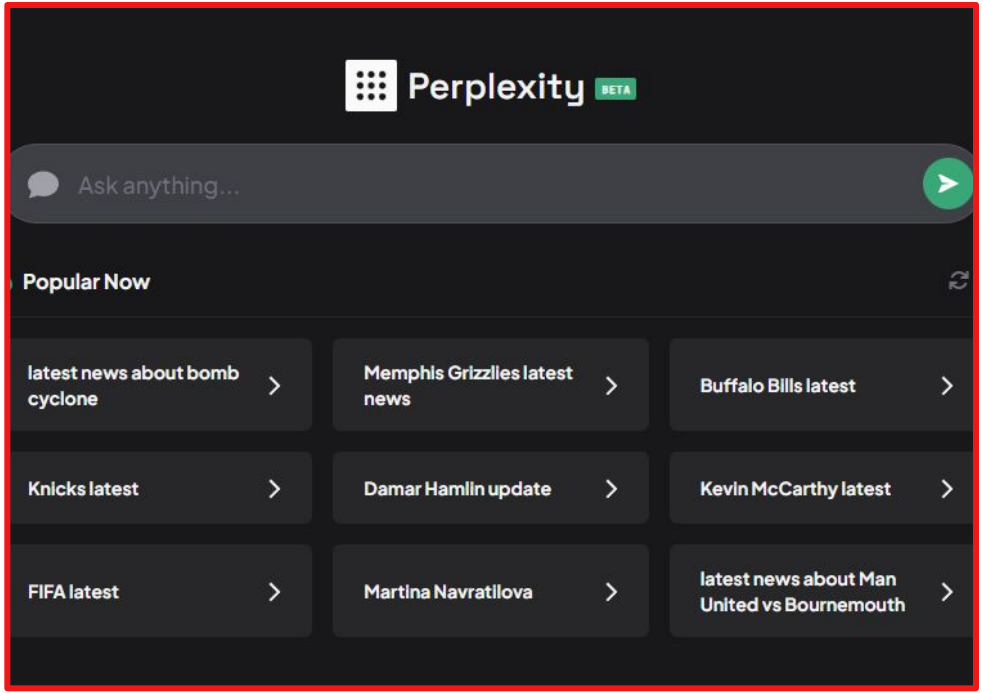
AI tools for academic research - General AI chatbots

- Like ChatGPT, Google Bard is also a general AI chatbot.
- But, google bard has access to all actual informations.
- It's totally free, and it's still in experimentation.



AI tools for academic research - General AI chatbots

- Perplexity, unlike ChatGPT and BARD, gives us the source of information.



AI tools for academic research - Academic AI chatbots

- Scite can get answers from million of papers, do literature review and provide smart citations.



AI tools for academic research - Academic AI chatbots

- Chatting with your docs with chatDoc !
- Quickly extract, locate and summarize information.
- For me, chatDoc is the best AI tool for researchers.



ChatDOC Tutorial



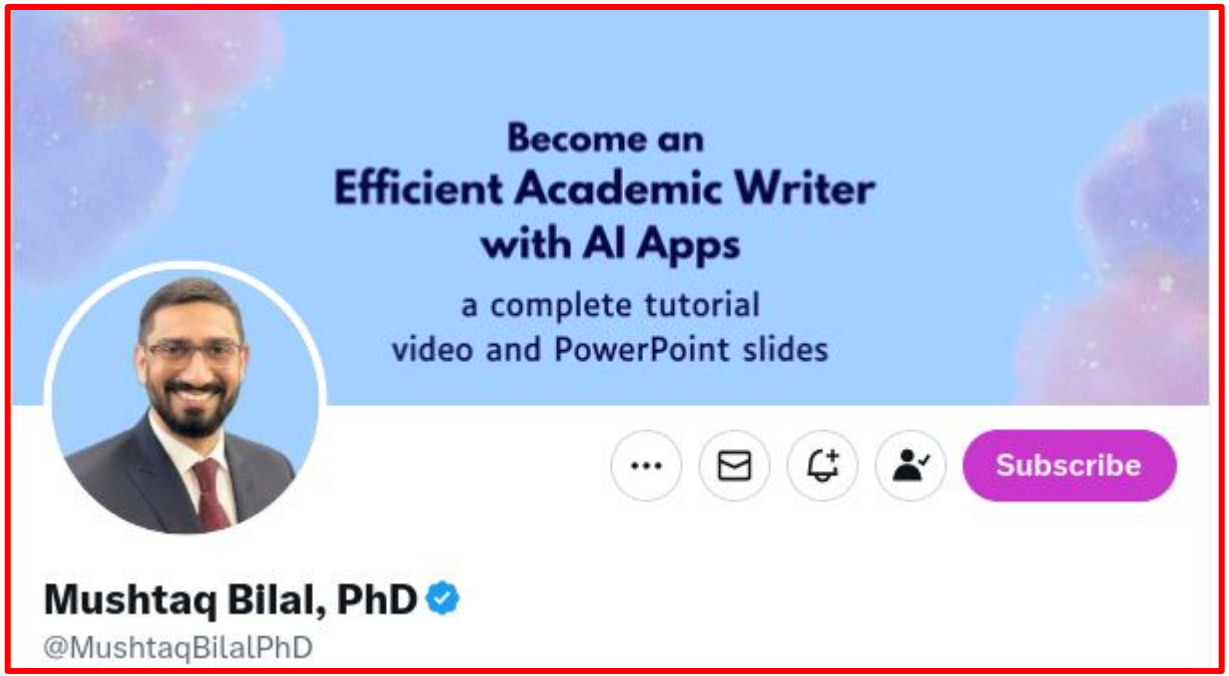
https://shorturl.at/tuIN3

Workshop : incorporate AI into your research process



https://shorturl.at/ilFGP

Bonus



Follow this account on X (ex.Twitter)

Bonus



Follow this account on Linkedin

Thank you all ^^